Analytical Chemistry 1 (125101)

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Framework of this course

❖ Introduction (L1)

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

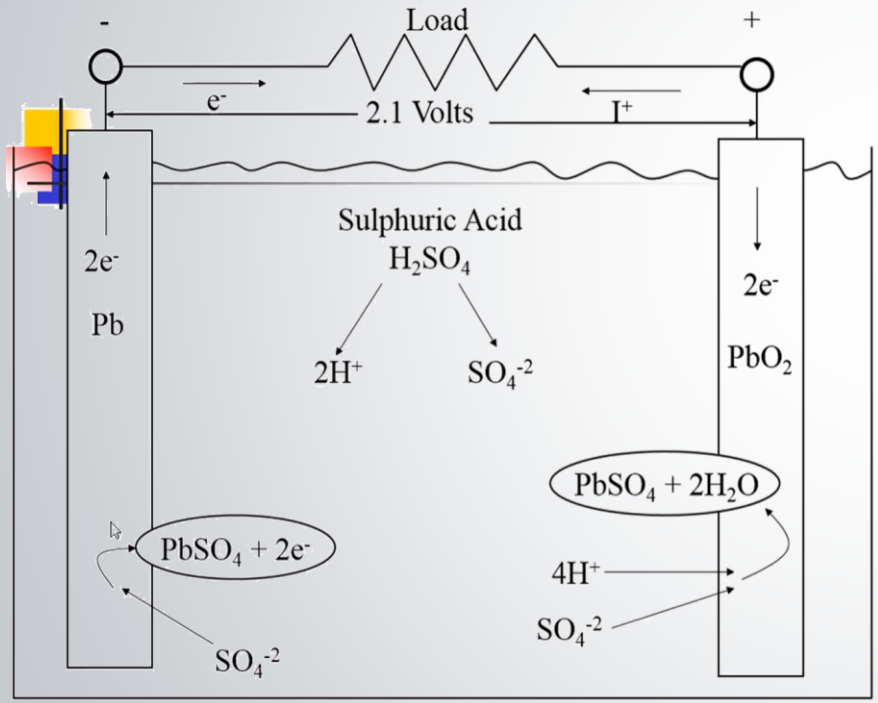
2.6 Gravimetric analysis (L12)

❖ Summary (L13)

2.5 Electroanalytical techniques

Electrolysis: A process in which external potential is used for forcing a reaction.

Example: Lead-acid battery



Discharging

Net reaction:

$$Pb(s) + PbO_2(s) + 2H_2SO_4 \rightleftarrows 2PbSO_4 + 2H_2O$$

$$\textit{Anode}: Pb(s) + SO_4^{2-} \rightleftarrows PbSO_4\,(s) + 2e \quad E^o = -Eo\,(PbSO_4 \mid Pb) = 0.355\ V$$

$$\textit{Cathode}: PbO_2(s) + 4H^+ + SO_4^{2-} + 2e \rightleftarrows PbSO_4 + 2H_2O \quad Eo = 1.685\ V$$

Electrolysis: A process in which external potential is used for forcing a reaction.

Charging a battery

$$2PbSO_4 + 2H_2O \rightleftarrows PbO_2(s) + Pb(s) + 2H_2SO_4 \quad E^o = -1.458\ V$$

← **Spontaneous direction**

Applying external potential higher than 1.5 V

<u>Reverses the reaction</u> and produce PbO_2 and Pb precipitates.

2.5 Electroanalytical techniques

Overview

2.5-1 Voltammetry analysis

2.5-2 Electrogravimetric analysis

2.5-3 Coulometric analysis

2.5-1 Voltammetry

Electroanalytical methods in which the analyte conc. is found by measuring the current as a function of applied potential under the conditions of conc. polarization. Only a minor portion of the analyte is consumed, and its conc. is thought to be unchanged.

The experimental setup

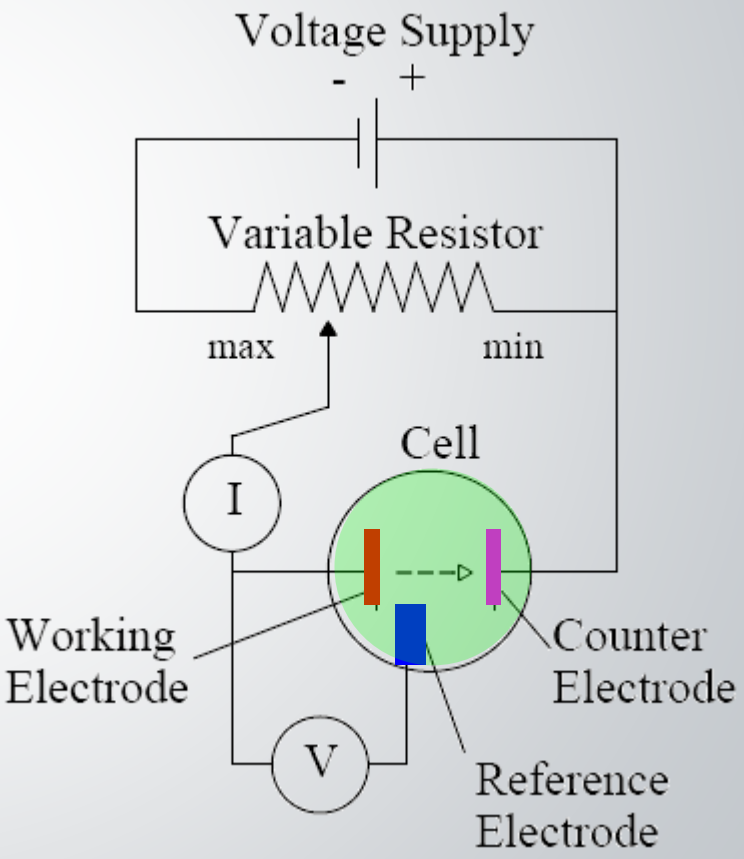
Three electrodes in analyte solution

(1) Working electrode:
of variable potential

(2) Reference electrode:
of constant potential.
No current passes.

(3) Counter electrode: Pt or Hg
for passing current to/from
the working electrode.

Supporting electrolyte:
inert electrolyte, in excess.

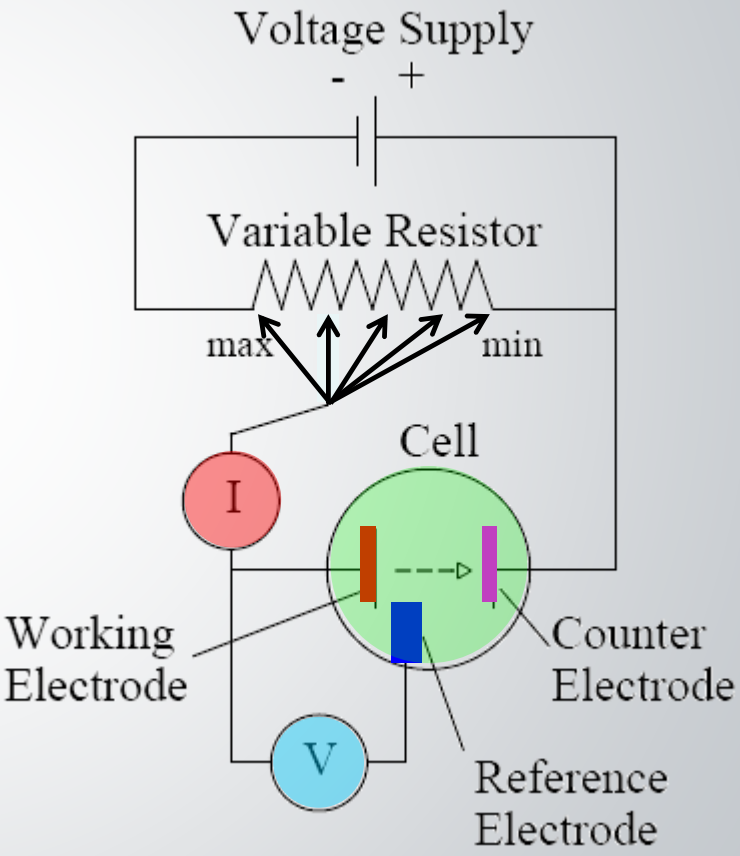


The experimental setup

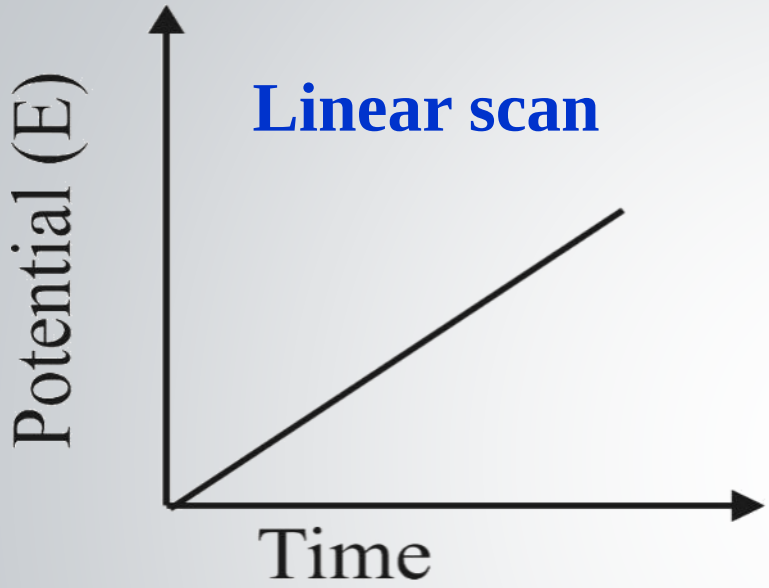
Variable voltage supply

Ampermeter

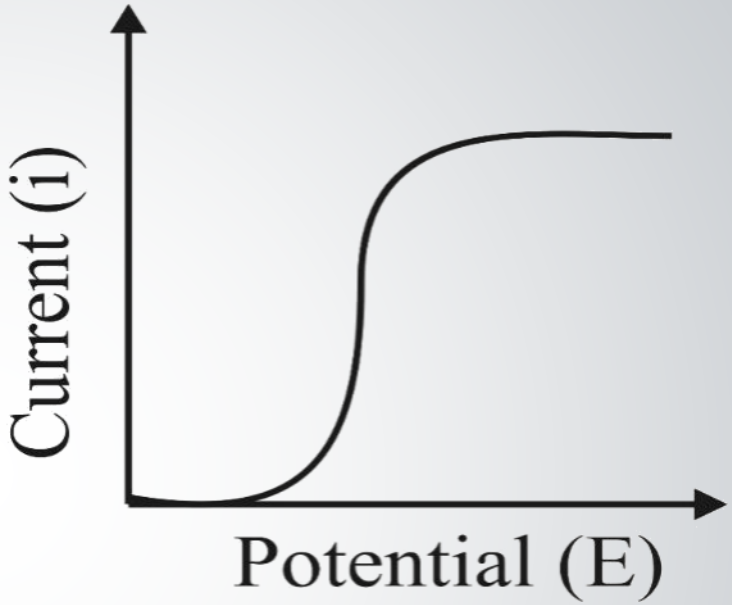
Potentiometer



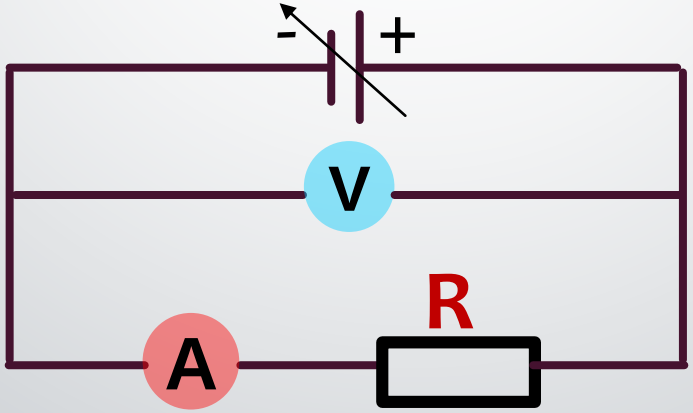
Applied potential as a function of time



Measured current as a function of potential



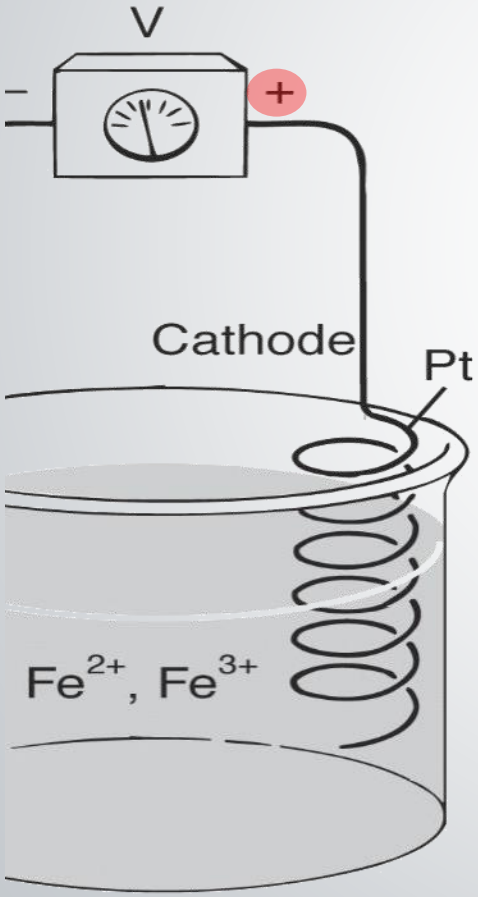
Variable voltage source

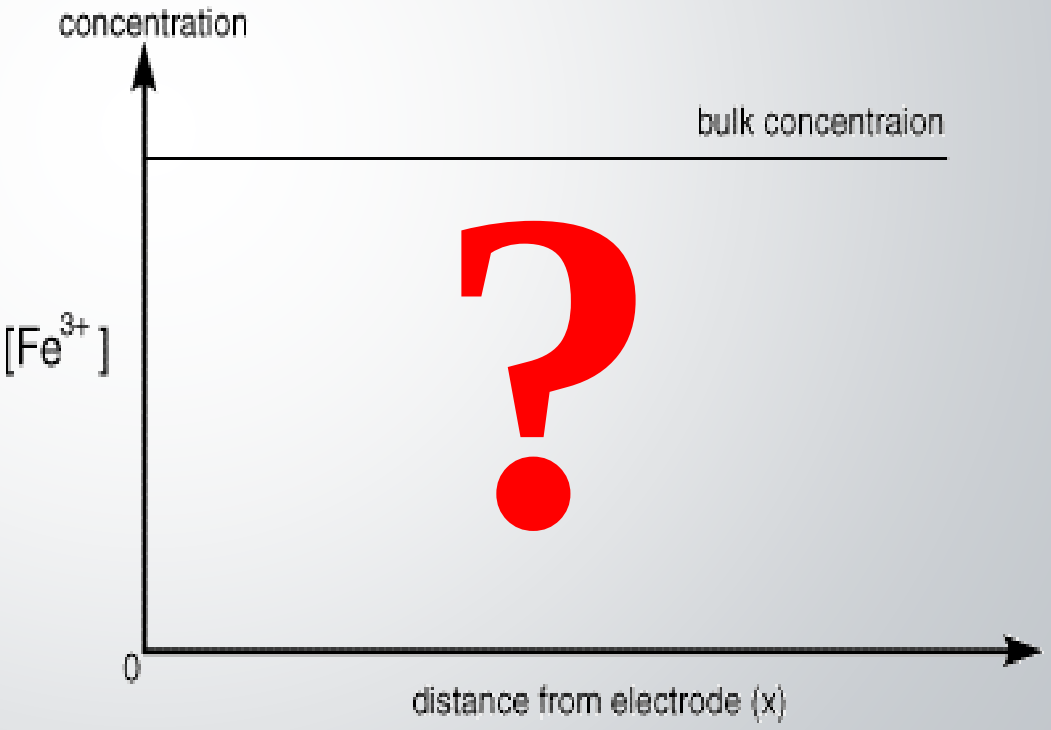


Circuit diagram

The concentrations as a function of distance from working electrode

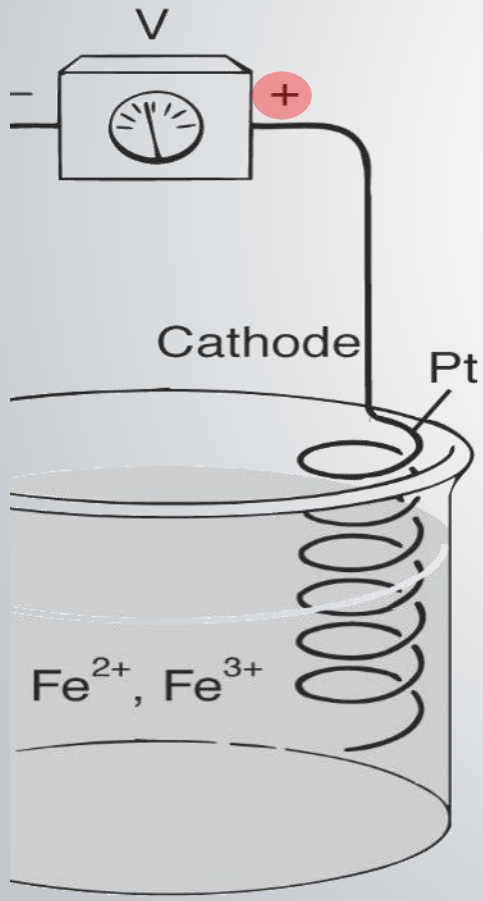
The reaction of electrode surface: $Fe^{3+} + e^{-} \rightarrow Fe^{2+}$





The concentrations as a function of distance from working electrode

Three ways to help the ions reach the surface of working electrode:

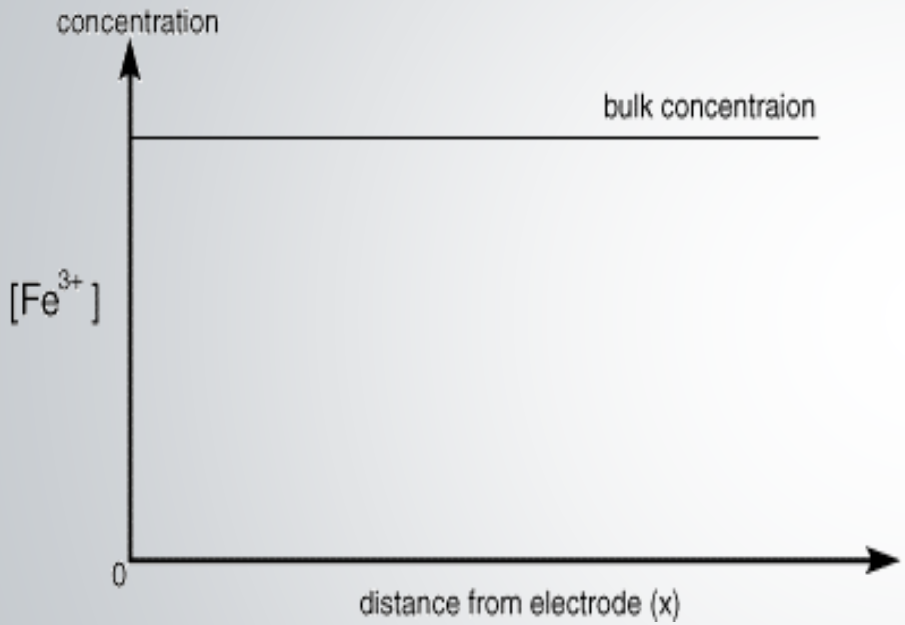


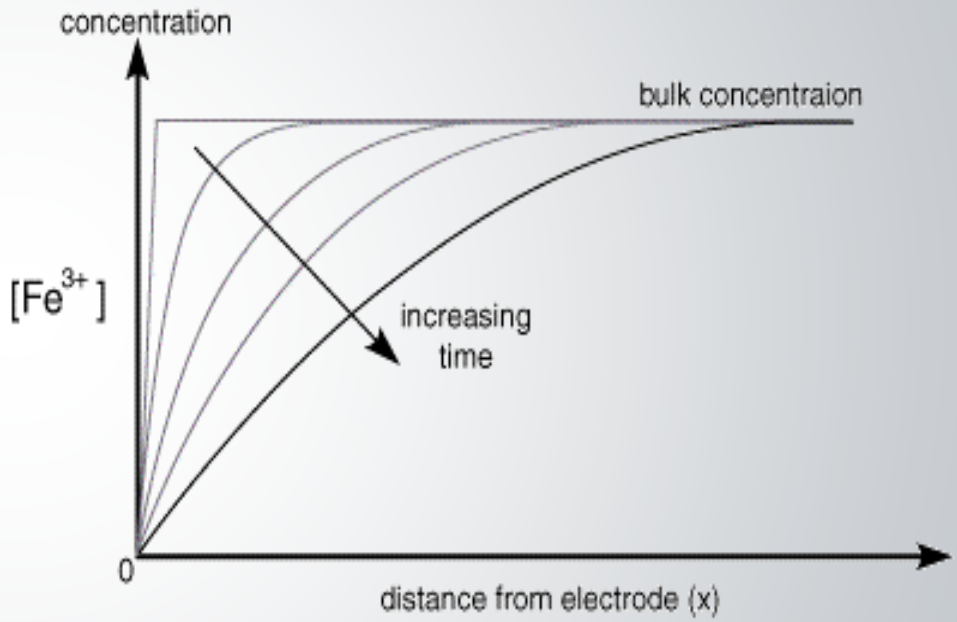
1) Diffusion through a conc. gradient

2) Convection through by physical means
   e.g. stirring or boiling…

3) Migration by a charged surface
   electrostatic attraction or repulsion

The reaction of electrode surface: $Fe^{3+} + e^{-} \rightarrow Fe^{2+}$





Before application of potential

After application of potential

Voltammetry

In voltammetry, the current is determined by rate of ion migration to working electrode surfaces.

-The selectivity to analyte is obtained by setting the potential on working electrode.

- Ions are attracted to the working electrode, where electroactive ions undergo redox reactions.

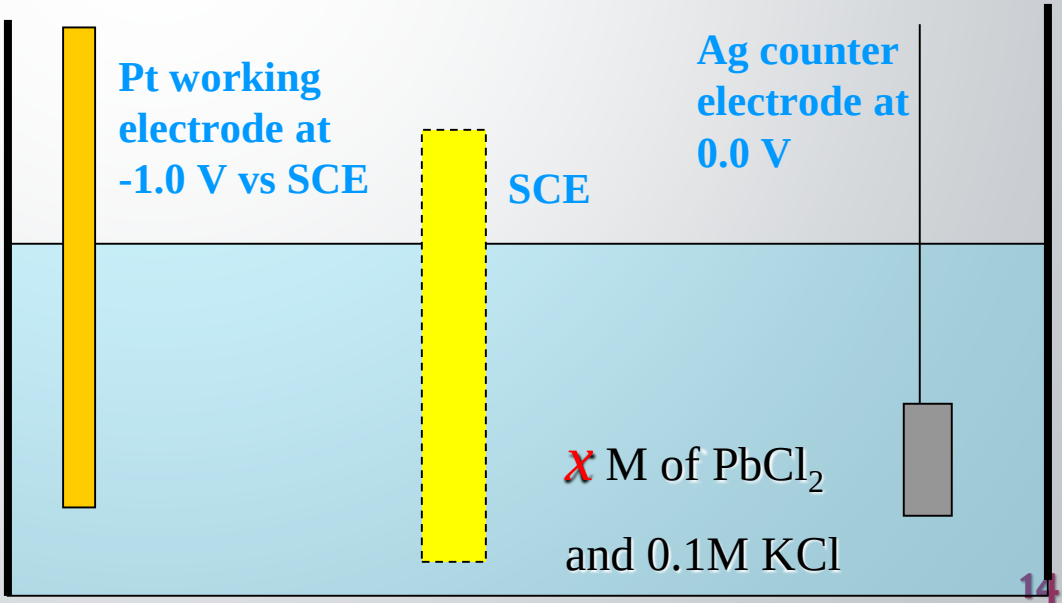
- The resulted current is proportional to the analyte conc.

Example:

Voltammetric analysis of Pb^{2+}

$Pb^{2+} + 2e^- \longrightarrow Pb \quad E^O = -0.13\ V$

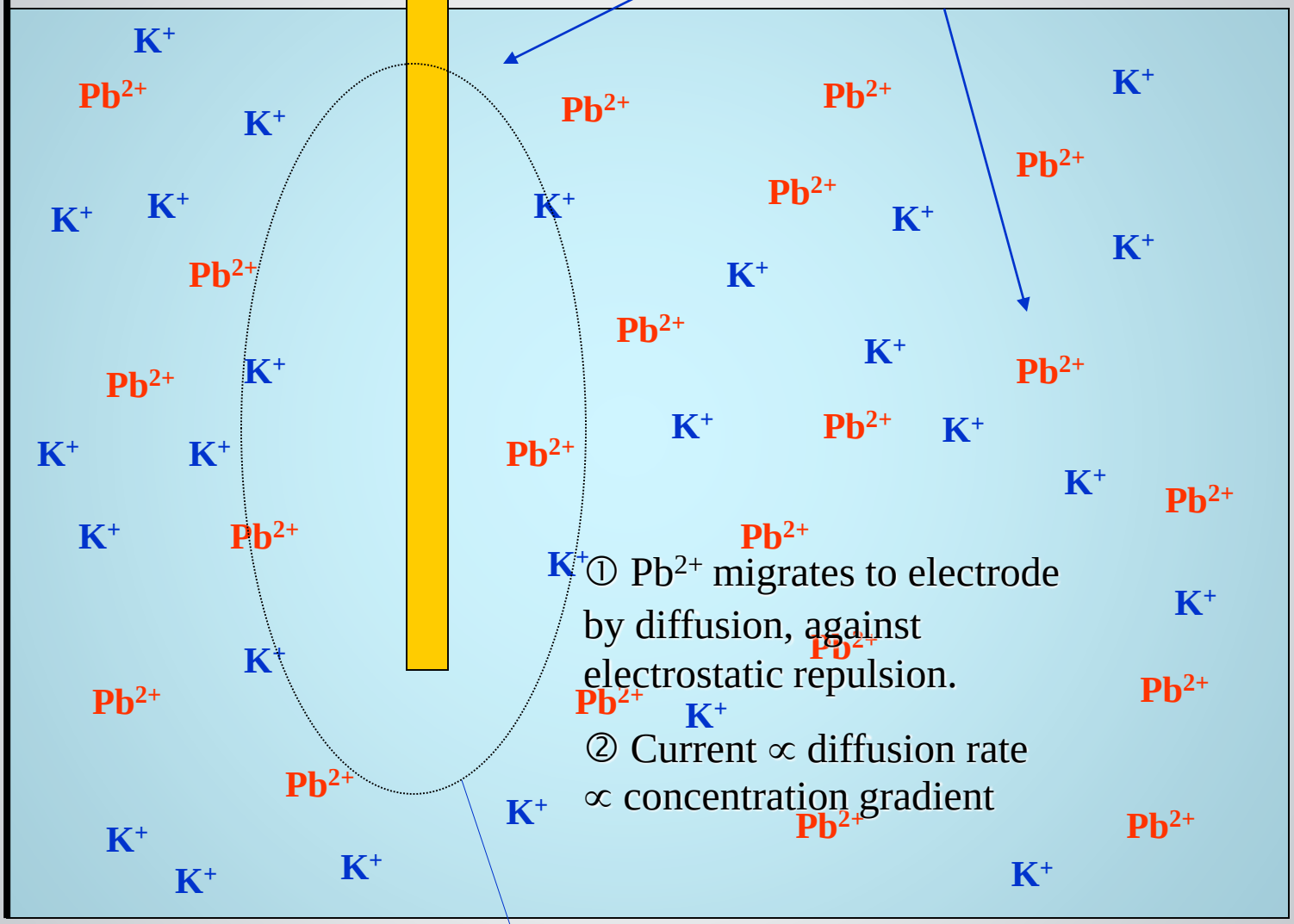
$K^+ + e^- \longrightarrow K \quad E^O = -2.93\ V$



$Pb^{2+} + 2e^- \longrightarrow Pb$

-1.0 V vs SCE

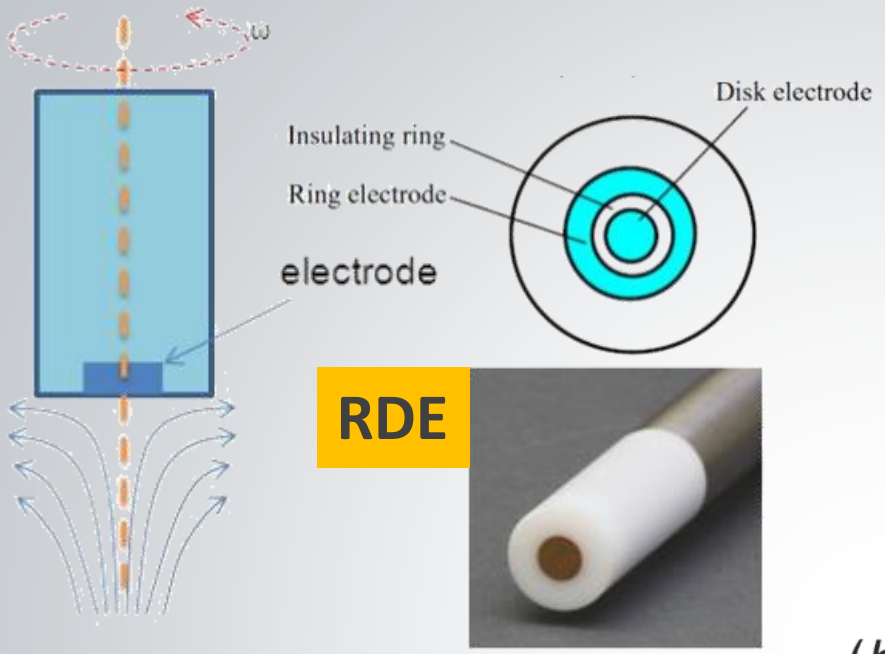
Concentration gradient is formed between the electrode and solution bulk



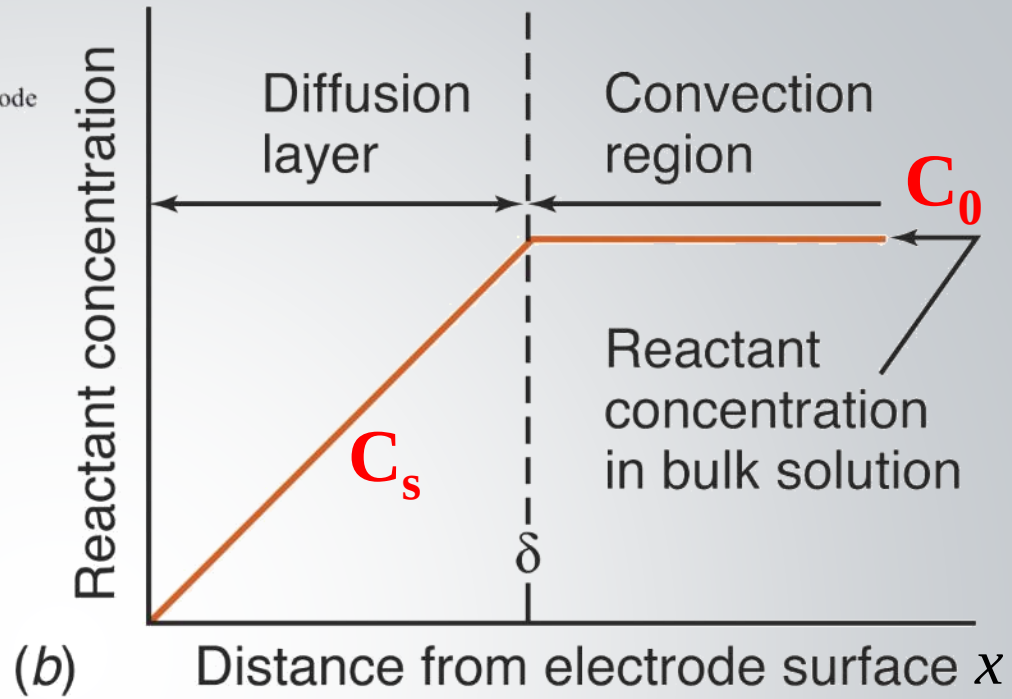
$Pb^{2+} + 2e^- \rightarrow Pb \quad E^O = -0.13$ V

$K^+ + e^- \rightarrow K \quad E^O = -2.93$ V

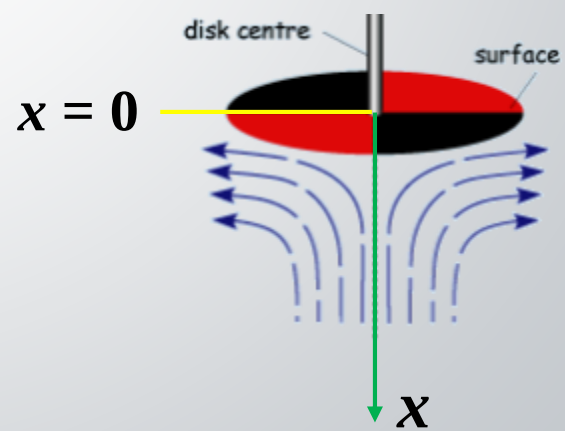
K^+ layer around the electrode repulses Pb^{2+} ions and almost stops their migration



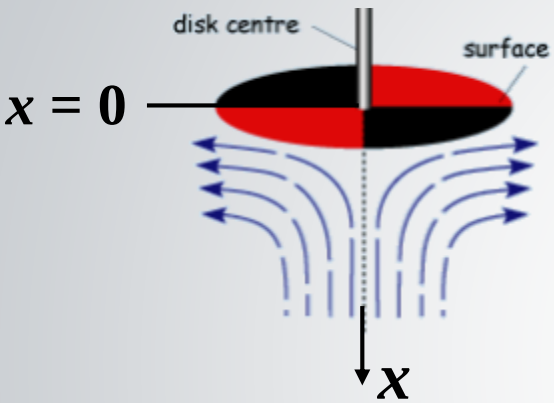
RDE: rotating disk electrode

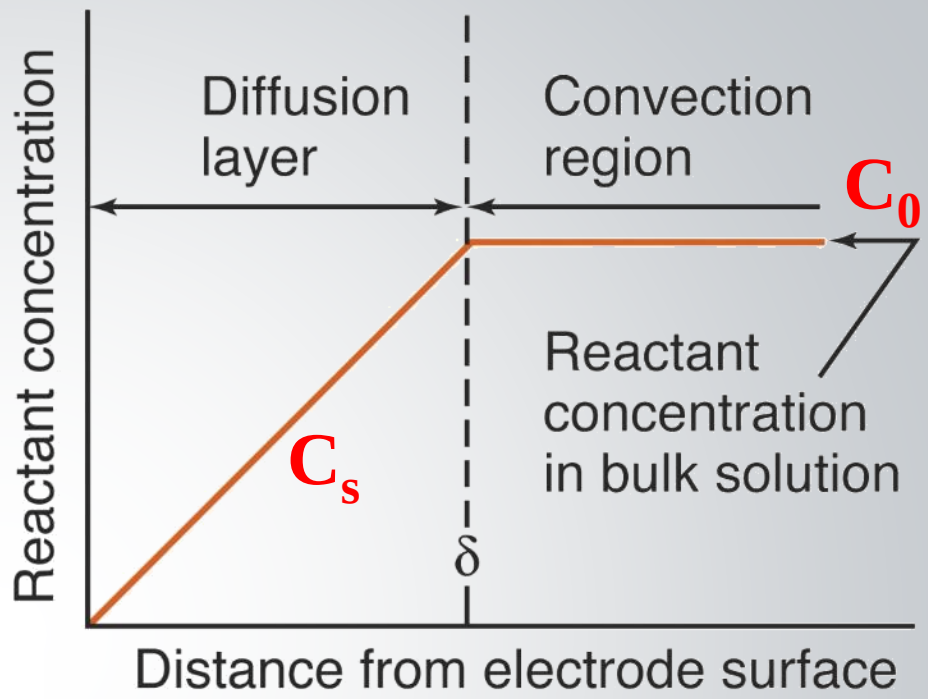


- If the potential is great enough, the reaction is fast, and the C_s at $x = 0$ is nearly 0.
- The conc. gradient is shown in the Fig. (0 μm < x < 100 μm) by diffusion alone.



Voltammetry





- Diffusion current is governed by the rate of analyte diffuses to working electrode surface.
- Current ∝ **diffusion rate** ∝ **($C_0 - C_s$) ~ C_0**

"∝" means "in proportion to"

Current is proportional to the C_0

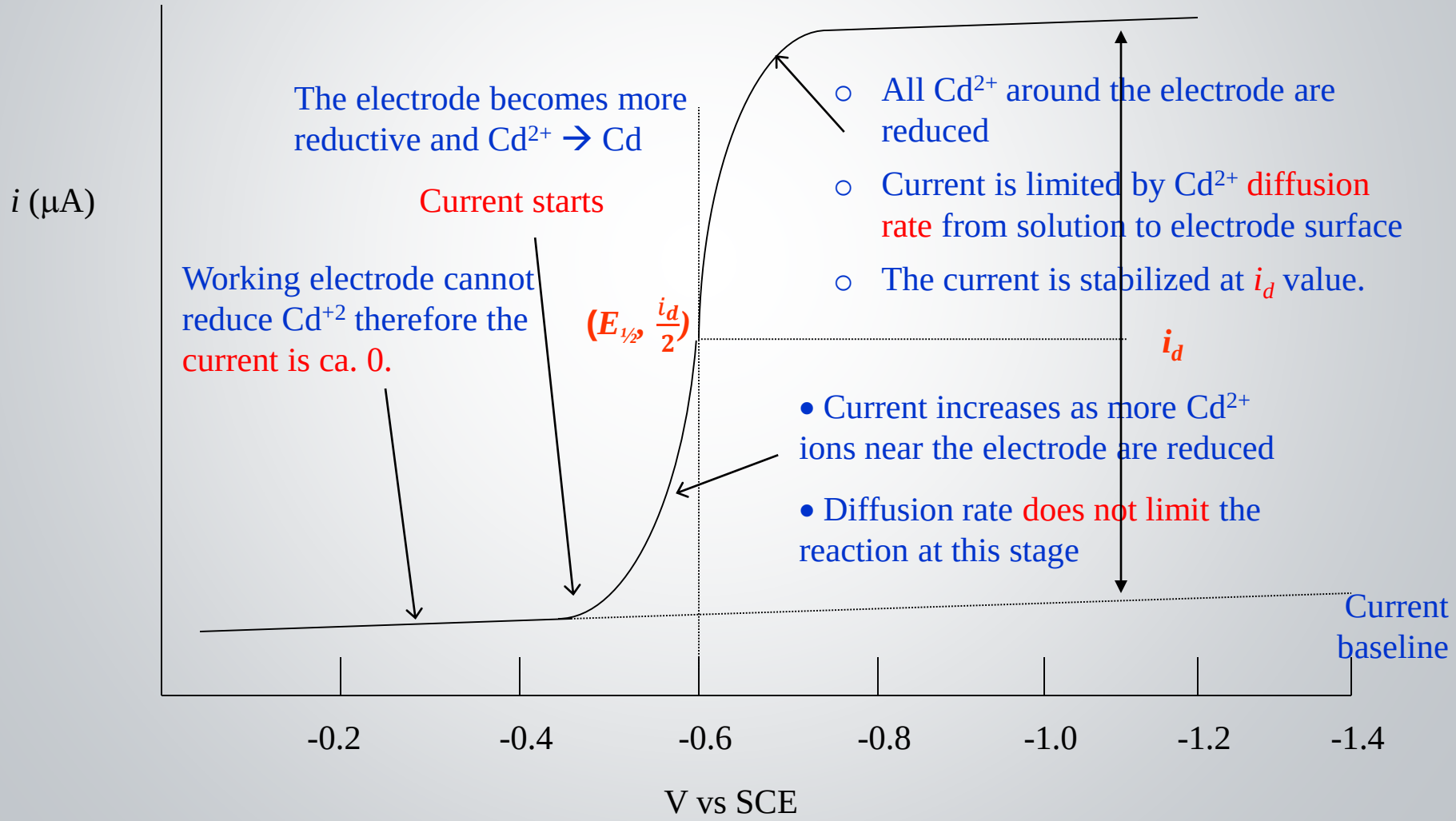
This is the basis for quantitative analysis by Voltammetry

Current as a function of applied potential - qualitative

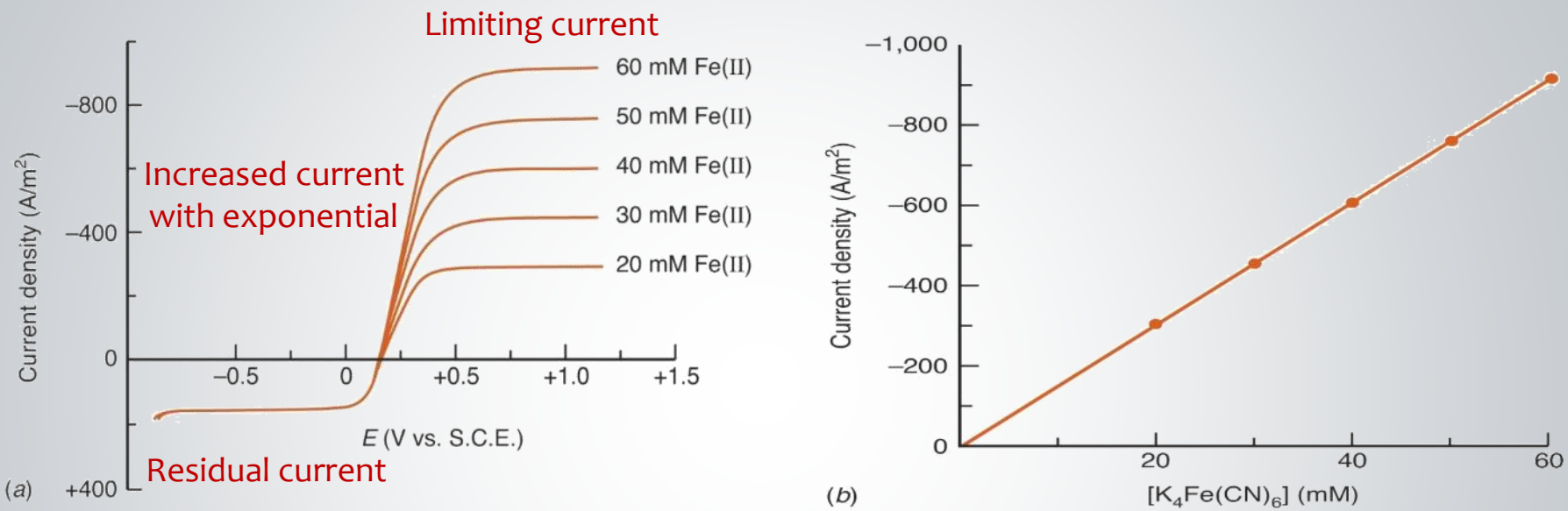
0.001 M Cd^{2+} in 0.1 M KNO_3 supporting electrolyte

$Cd^{2+} + 2e^- \rightarrow Cd \quad E^O = -0.402$ V

$K^+ + e^- \rightarrow K \quad E^O = -2.93$ V



Voltammetry

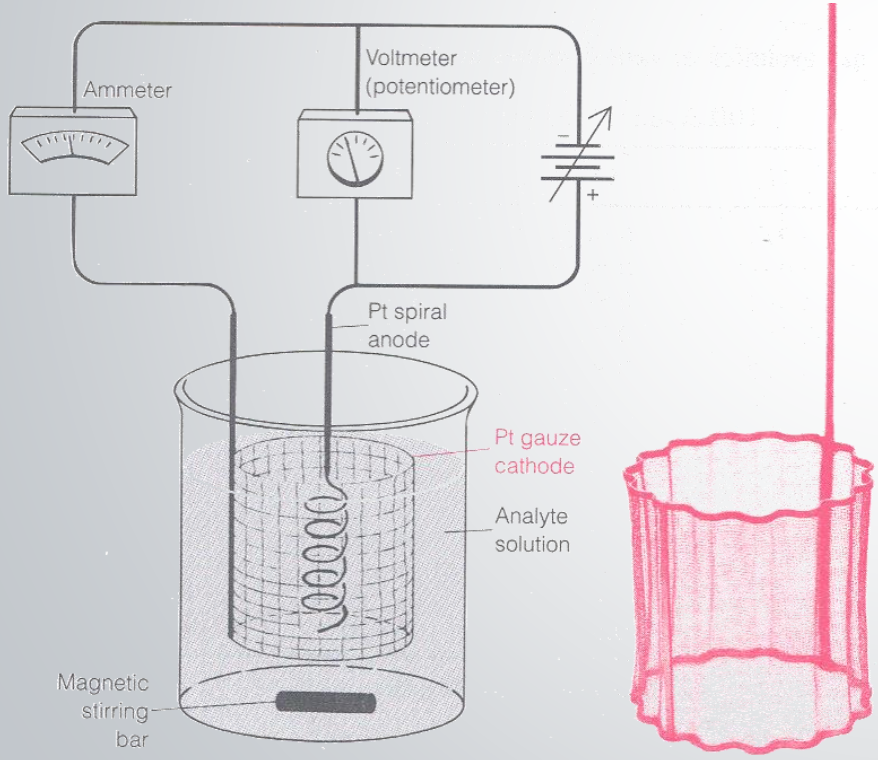


- **The relation between current and voltage during the electrochemical processes.**

- **The current intensity is proportional to the conc. of Fe^{2+}.**

2.5-2 Electrogravimetric analysis

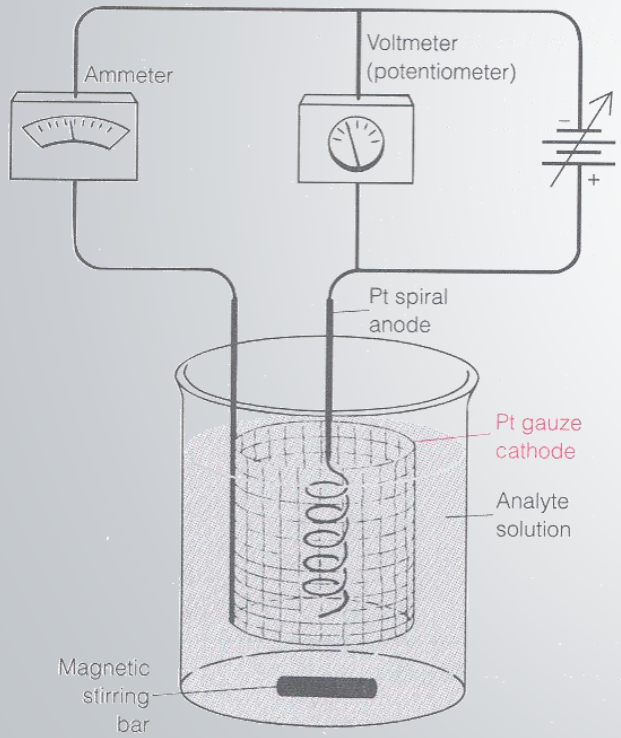
Quantitative analysis based on precipitating all the analyte on an electrode and the electrode is weighing before and after deposition.



How do we know that precipitation is completed?

- Disappearance of color
- No deposition on freshly exposed electrode surface
- Qualitative analysis of the solution

2.5-2 Electrogravimetric analysis
(Electrolytic precipitation)



Cathode reaction:

$Cu^{2+} + 2e \longrightarrow Cu\ (s)\quad E^o = 0.342\ V$

Anode reaction:

$2H_2O \longrightarrow O_2\ (g) + 4H^+ + 4e\quad E = -1.23\ V$

Net reaction:

$2Cu^{2+} + 2H_2O \longrightarrow Cu\ (s) + O_2\ (g) + 4H^+$

Nernst equation:

The required potential $E_{applied}$

$$= E^o(Cu^{2+}/Cu) + E\ (H_2O/O_2) - \frac{0.05916}{4}\log\frac{[H^+]^4 p(O_2)}{[Cu^{2+}]^2}$$

2.5-3 Coulometric analysis

The quantitative analysis based on reacting **all analyte** and measuring the **total charges consumed.**

A. Constant potential

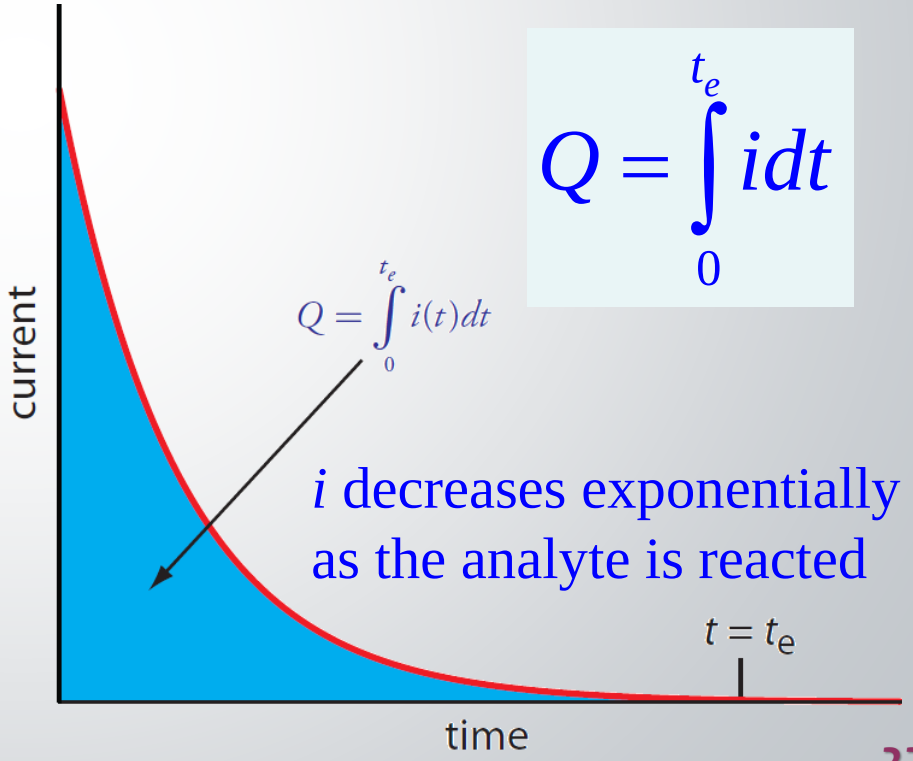
Drawback:

Time consuming

Solutions:

- Small sample volume
- Large electrode area
- Efficient mixing
- Constant current method

$$Q = \int_0^{t_e} i dt$$



i decreases exponentially as the analyte is reacted

Selecting the constant potential

Example:

Analysis of Cu^{2+} by their precipitation on Pt electrode

$$Cu^{2+}_{(aq)} + 2e^- \rightleftarrows Cu_{(s)} \qquad E^o = +0.34\ V$$

The electrode gives the electrons

The electrode must be sufficiently negative to deliver electrons to Cu ions

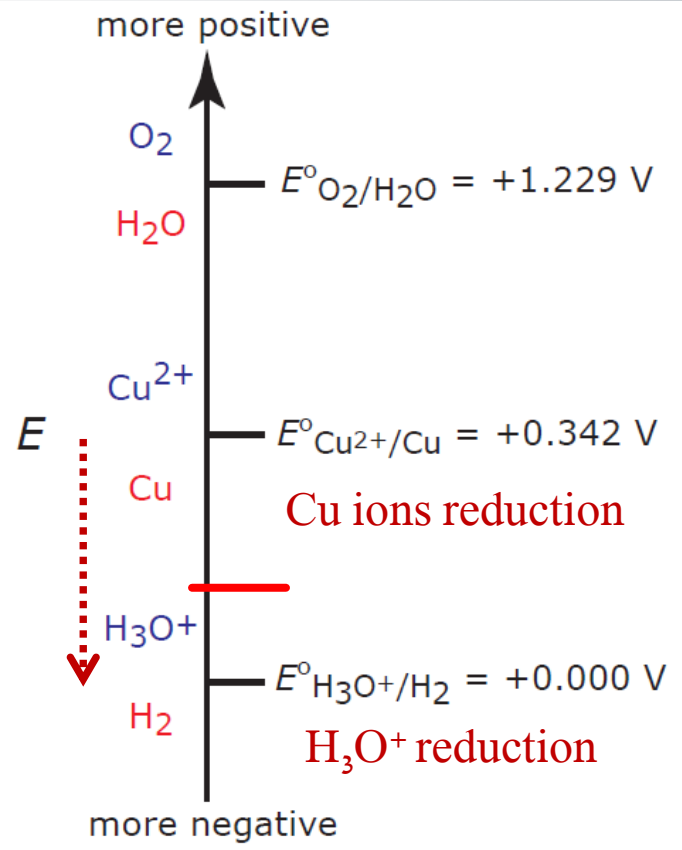
Q:

How negative may the electrode be?

A:

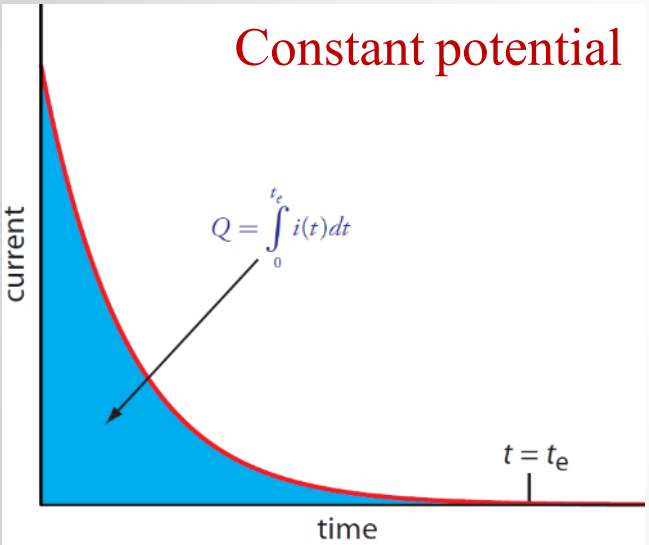
Till the hydronium ions are reduced to H_2.

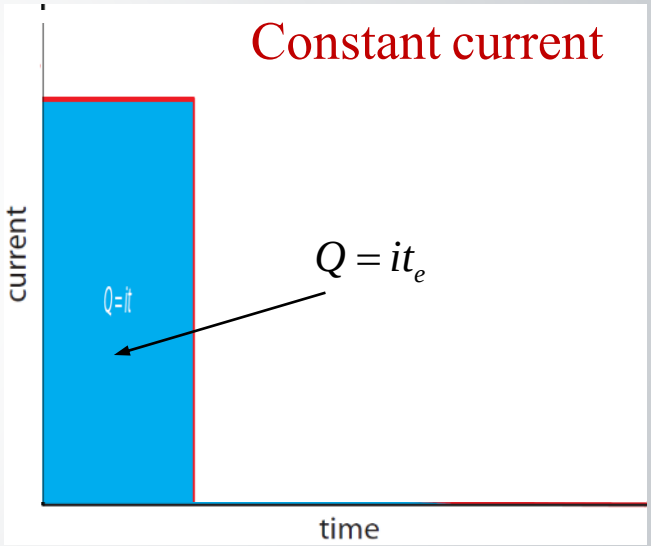
At this point the current is not solely attributed to the reduction of Cu ions.



B. Constant current

Increasing the potential with time, so that the current remains constant





Advantages:

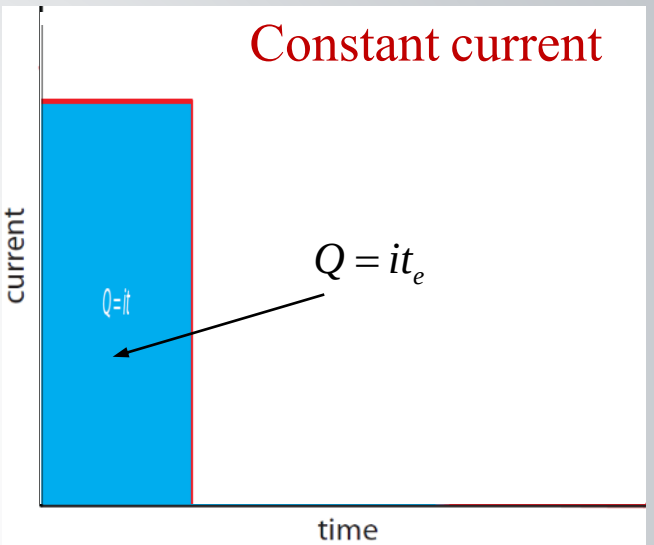
- Shorter analysis time
- Simpler integration

Formula

$$Q = I*t = n_e*F$$

B. Constant current

Q — the amount of charge
I — current
t — time
n_e — moles of electron participating in the chemical reaction
F — Faraday constant
9.6485309 $\times 10^4$ Coulomb/ mol



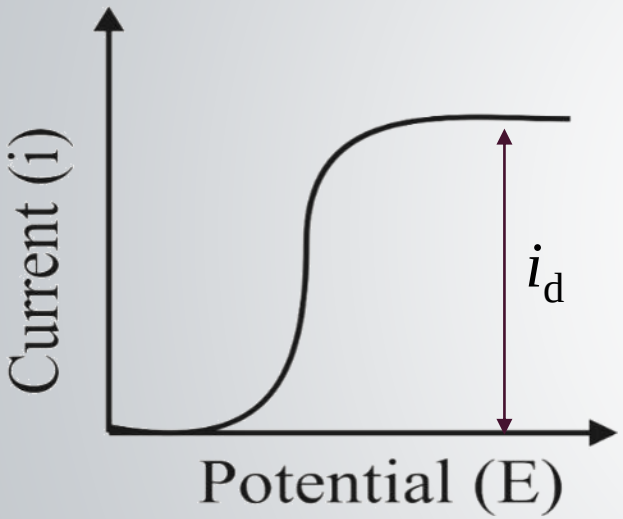
Formula

$$\mathbf{Q = I*t = n_e*F}$$

$$\text{Moles of } e^- = \frac{q(\text{Coulomb})}{F} = \frac{I(\text{Amperes})\cdot t(\text{seconds})}{F}$$

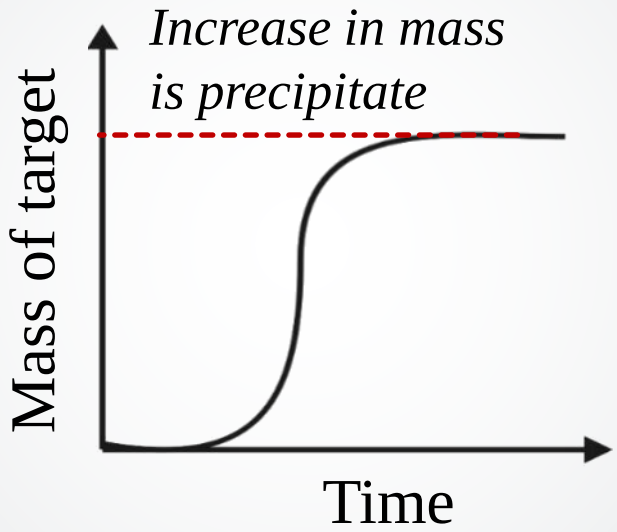
Comparisons with different electroanalysis methods

Voltammetry



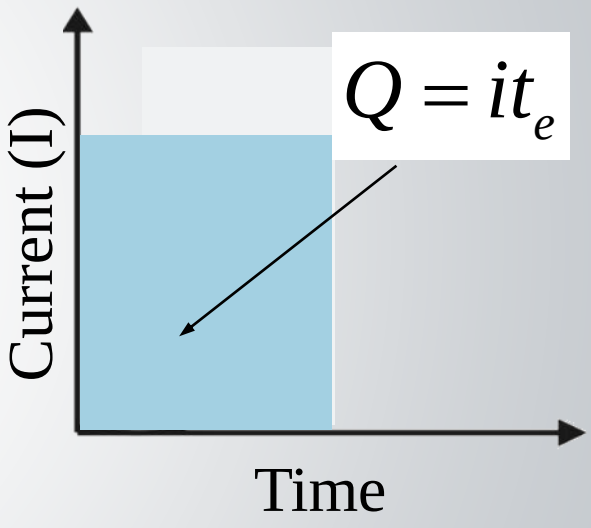
Measured current as a function of potential

Electrogravimetry



Measured precipitate weight

Coulometry



Measured current as a function of time

Electroanalytical techniques

1. Voltammetry analysis

Based on measuring the current intensity under the conditions of concentration polarization.

2. Electrogravimetric analysis

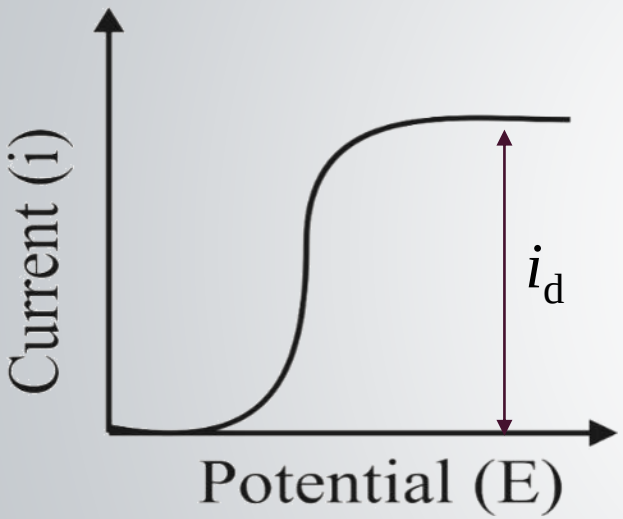
Based on the analyte is deposited as a solid on the electrode. The increase in the electrode's weight is a direct measure of analyte amount in the solution.

3. Coulometric analysis

is based on counting the number of electrons that participate in the chemical reaction.

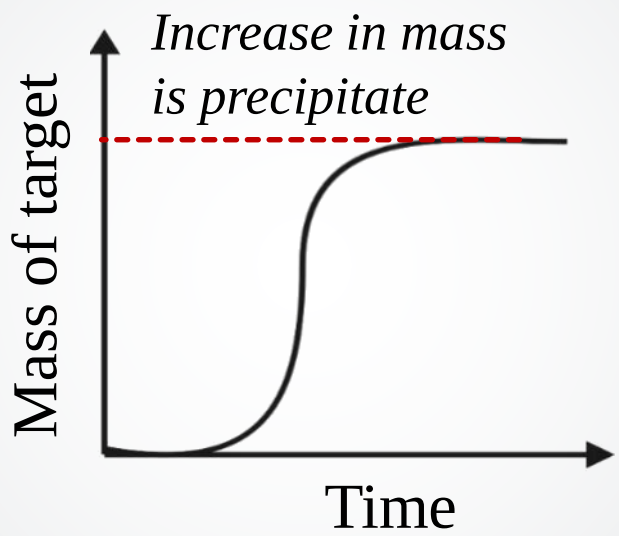
Comparisons with different electroanalysis methods

Voltammetry



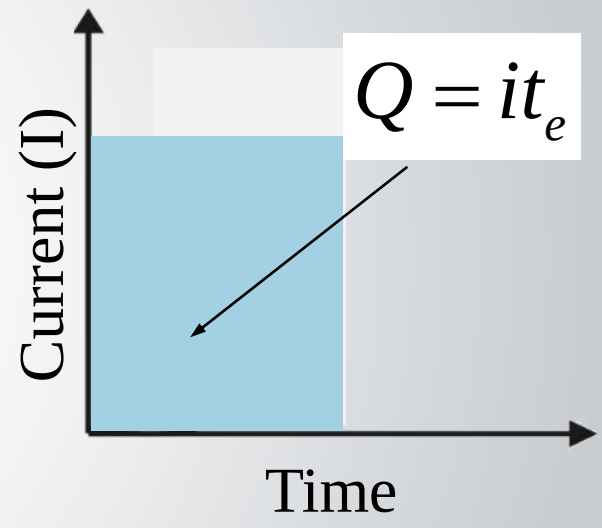
Measured current as a function of potential

Electrogravimetry



Measured precipitate weight

Coulometry



Measured current as a function of time

All these is about quantitative analysis, how about qualitative (selective)?